

Extra Practice Multiplying and Dividing Rational Expressions

MCR3U

1) Multiply the following rational expressions. Simplify and state restrictions.

a) $\frac{x+4}{x} \cdot \frac{x^2}{x^2+5x+4}$

b) $\frac{x^2+10x+16}{5x-10} \cdot \frac{x-2}{x^2+9x+8}$

c) $\frac{2x^2-10x}{x^2-9x+20} \cdot \frac{x^2-8x+16}{4x^2}$

d) $\frac{2x+4}{x+4} \cdot \frac{5x^2+21x+4}{10x+2}$

2) Divide the following rational expressions. Simplify and state restrictions.

a) $\frac{x^2-5x+6}{5} \div \frac{x-3}{15}$

b) $\frac{x^2-5x+6}{8x^2+24x} \div \frac{x-2}{4x+12}$

c) $\frac{x^2-2x+1}{x+1} \div \frac{x^2-1}{x+1}$

d) $\frac{x^2+7x+12}{x^2+3x-10} \div \frac{x^2-x-20}{x^2-25}$